

Chapter 9, Problem 1PP

Problem

Given the sinusoid $45 \cos(5 \pi t + 36^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Step-by-step solution

Step 1 of 4

Write the given sinusoidal function.

$$45 \cos(5 \pi t + 36^\circ)$$

Write the standard expression for the sinusoidal function.

$$A \cos(\omega t + \phi)$$

Here,

A is the amplitude of the sinusoid

ω is the angular frequency in rad/s

ϕ is the phase angle in degrees.

Step 2 of 4

Compare the given sinusoidal function with standard function.

The amplitude of the sinusoid is $\boxed{45}$.

The phase angle of the sinusoid is $\boxed{36^\circ}$.

Determine the angular frequency, ω .

$$\begin{aligned}\omega &= 5\pi \\ &= 15.708 \text{ rad/s}\end{aligned}$$

Therefore, the angular frequency, ω is $\boxed{15.708 \text{ rad/s}}$.

Step 3 of 4

Determine the time period of the sinusoid.

$$\begin{aligned}T &= \frac{2\pi}{\omega} \\ &= \frac{2\pi}{5\pi} \\ &= 0.4 \\ &= 400 \text{ ms}\end{aligned}$$

Therefore, the time period is $\boxed{400 \text{ ms}}$.

Step 4 of 4

Calculate the frequency of sinusoid.

$$\begin{aligned}f &= \frac{1}{T} \\&= \frac{1}{0.4} \\&= 2.5 \text{ Hz}\end{aligned}$$

Therefore, the frequency is 2.5 Hz.